

ZENELAIT HRMS FLOWCHARTS

PACKAGE-WISE ROLE WORKFLOW MATRIX & TRANSACTION PATHWAYS

An exhaustive technical document detailing every functional package and endpoint within Zenelait HRMS. Maps each of the three enterprise user roles (Super Admin, HR Admin, and Employee) to their corresponding frontend client views, backend API controllers, and relational database schemas.

ROLES COVERED

- SUPERADMIN - Systems administrator, tenant provider & auditor.
- ADMIN (HR Admin) - Tenant workforce manager, recruiter, payroll & project lead.
- EMPLOYEE (Staff) - Self-service timekeeper, claimant, ticket requester & learner.

TECHNICAL PACKAGES INTEGRATED

- Core Services: Auth, Onboarding, Roster, Exit, Dashboard
- Logistics & Time: Attendance, Timesheets, Leaves, Travel, Expenses
- Business & Operations: Projects, Sprints, Worklogs, Payroll, Assets, Tickets, Invoices
- Corporate Growth: Recruitment, Appraisals, Course Learning, Announcements

TABLE OF CONTENTS & SYSTEM LEGEND

REPORT INDEX & VISUAL DATA FLOW MAPPING GUIDES

REPORT PAGES INDEX

- Page 1** — Cover Page & Document Purpose
- Page 2** — Table of Contents & Flowchart Legend Guide
- Page 3** — Super Admin Role Flowchart (System Operations & Organization Control)
- Page 4** — HR Admin Role Flowchart - Part I (Onboarding, Directory, & Recruitment)
- Page 5** — HR Admin Role Flowchart - Part II (Attendance, Leaves, & Sprint Projects)
- Page 6** — HR Admin Role Flowchart - Part III (Expenses, Payroll, Invoices, & Assets)
- Page 7** — Employee Role Flowchart - Part I (Auth, Dashboards, Clocking, & Leaves)
- Page 8** — Employee Role Flowchart - Part II (Sprints, Travel, Appraisals, & Helpdesk)
- Page 9** — Relational Architecture Matrix & Comprehensive Package Schema Mapping

FLOWCHART COLOR LEGEND & PIPELINE DIRECTIONS

ROLE / SYSTEM INITIATOR	User triggers actions (credentials entry, form submissions) inside their specific account scope.
FRONTEND CLIENT VIEW	The React/Vite dashboard client UI routes, pages, and modular layouts accessed by users.
BACKEND API CONTROLLER	The Java Spring Boot REST controllers mapping the incoming JSON requests and executing validations.
DATABASE RELATIONAL ENTITY	MySQL / TiDB tables storing transactional ledgers, organization mappings, logs, and rosters.

DOCUMENT READ INSTRUCTIONS:

1. All flowcharts are mapped horizontally from Left to Right: User Context -> Frontend Client View -> Spring Boot REST Controller -> Relational Database Table.
2. Connections indicate REST API operations (GET, POST, PUT, DELETE) and events dispatched to Kafka topics.
3. Every controller class from the `com.zenelait.hrms.controller` Java package is accounted for in the flow diagrams.

SUPER ADMIN WORKFLOW FLOWCHART

SYSTEM-LEVEL TENANT PROVISIONING AND CORE AUDIT LEDGER MATRIX



HR ADMIN WORKFLOW FLOWCHART - PART I

ONBOARDING LOGISTICS, EMPLOYEE ROSTERS, AND CANDIDATE PIPELINES



HR ADMIN WORKFLOW FLOWCHART - PART II

WORKFORCE TIMEKEEPING, LEAVE APPROVALS, AND SPRINTS / PROJECTS



HR ADMIN WORKFLOW FLOWCHART - PART III

EXPENSE LEDGERS, ENTERPRISE PAYROLL, INVOICING, AND HELP TICKET AUDITS



EMPLOYEE WORKFLOW FLOWCHART - PART I

PORTAL LOCK BYPASS, PERSONAL DASHBOARDS, WORKCLOCK LOGS, AND LEAVE APPLICATIONS



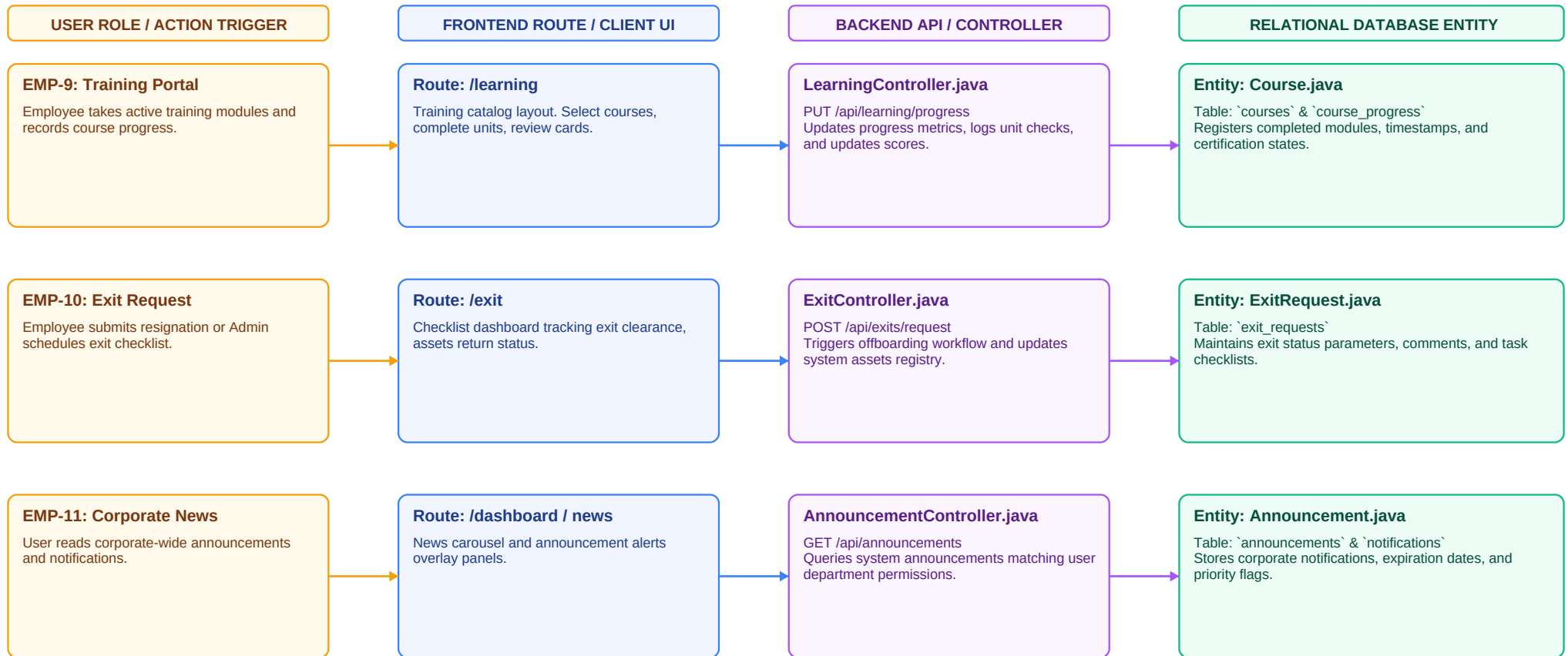
EMPLOYEE WORKFLOW FLOWCHART - PART II

SPRINT TICKETS, TRAVEL CLAIMS, KPI PERFORMANCE, HELPDESK TICKETS, AND TRAINING



REMAINING PACKAGES & ARCHITECTURE MATRIX

LEARNING SYNC, EXIT LOGISTICS, ANNOUNCEMENTS, AND SYSTEM WIDE DATA PATHS



CORE RELATIONAL SYSTEM CONSTRAINTS:

- Tenant Isolation: Every database transaction queries where `organization\_id = {user.org.id}` matching credentials.
- Asynchronous Event Brokerage: System operations audit events are sent to Kafka topics asynchronously, avoiding database lock delays.
- Redis Cache Synchronization: User session permission matrices are stored in memory and cleared on password resets or profile updates.